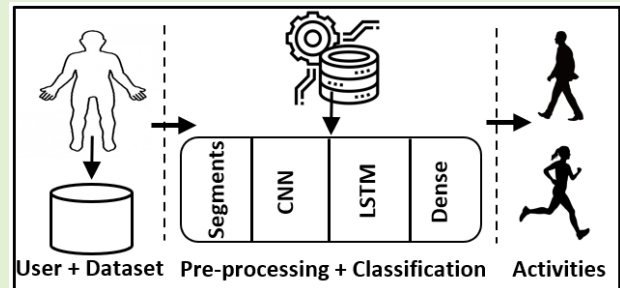


KNEE-HAR: A ConvLSTM-Based Lightweight Hybrid Deep-Learning Model for Human Activity Recognition With Knee Abnormalities Using Raw Physiological Sensor Data

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Abstract—Human activity recognition (HAR) is one of the most exploited areas in the field of artificial intelligence and wearable sensor technology. Most of the work on HAR classifies simple to complex human activities but does not manage to recognize the actions of an individual with distinct physical abnormalities. This article proposes an efficient and lightweight hybrid deep learning model (ConvLSTM) to classify human activities with knee abnormalities. A state-of-the-art publicly available physiological sensor-based HAR dataset with knee abnormalities is incorporated to validate the proposed framework. A minimal data preprocessing pipeline was also introduced to prepare the dataset for categorization without interfering with its raw characteristics. Further, optimal hyperparameters with an efficient time-step-based data segmentation method were developed for automatic feature engineering in spatial and temporal frequency domains. Moreover, incorporating regularization and low-cost mathematical functions resulted in a sparse and prominent feature space, improving performance by overcoming overfitting and optimizing computational times. The proposed model achieved the highest accuracy of 99.21% and an average accuracy of 99% in comparatively optimized computational time than the benchmark and state-of-the-art models.

Index Terms—Activity recognition, deep learning, feature engineering, knee abnormalities, patient rehabilitation, physiological sensors.



I. INTRODUCTION

THE field of healthcare has seen significant advances in the utilization of physiological sensors for numerous applications like—patient rehabilitation, elderly care, and sleep monitoring. These sensors play a crucial role in monitoring and estimating various physiological parameters, facilitating healthcare specialists to gather valuable insights about patients' health conditions. The incorporation of wearable devices like—smartwatches, fitness trackers, and medical-grade sensors has revolutionized healthcare applications by providing continuous and real-time monitoring of vital signals [1], [2], [3], [4]. The continuous expansion of physiological sensors holds great promise for improving healthcare facilities by facilitating different models with efficiency and

optimal performance. One such field incorporating wearable sensors and smart learning algorithms is human activity recognition (HAR) [5], [6], [7]. Human activity refers to any movement or physical action performed by an individual while engaged in tasks, recreational activities, or household responsibilities, and identifying these movements is known as HAR.

A. Problem Statement and Motivation

Machine- and deep-learning algorithms are used extensively for numerous tasks like object detection, classification, signal processing, and many more [8], [9], [10], [11]. Machine-learning or shallow learning models are widely utilized for classification tasks, employing a combination of input features and corresponding class labels. These models can understand complex relationships within the dataset features and can generate useful patterns for future predictions. However, the performance of machine-learning algorithms depends highly on manual feature engineering and intense data preprocessing. In contrast, deep learning exploits the potential of multilayer neural networks to extract hidden patterns from the collected

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data automatically and eliminates the necessity for manual feature engineering. Nonetheless, the efficient performance of deep-learning models needs high-quality datasets with numerous instances and optimized hyperparameter tuning to achieve optimal results.

Knee injuries are a prevalent worldwide concern affecting people across various age groups in their daily activities. The frequency of knee injuries can be attributed to multiple factors like—sports activities, accidents, falls, and many more. These injuries can significantly impact an individual's mobility, quality of life, and overall well-being [3], [12]. With the interface of HAR systems, rehabilitation of knee injuries can restore the affected knee's strength, flexibility, and functionality by facilitating the healthcare professional with a robot-based assistant for activity monitoring and personalized care. Subsequently, electromyography (EMG) sensors record electrical signals produced by muscles while performing an activity, giving valuable insights into muscle activation patterns and anomalies in individuals [12], [13], [14]. By leveraging the hierarchical feature engineering and complex relationship-capturing behavior of deep learning with the raw sensor reading of electrical activity of EMG sensors, this article proposes an efficient and lightweight ConvLSTM model for HAR with Knee Abnormalities.

The main contributions and novelties of this article are as follows.

- 1) An effective feature fusion pipeline of hybrid deep-learning architecture (ConvLSTM) is proposed for automatic feature engineering and enhanced knee abnormalities-based activity recognition on the raw physiological data generalized performance.
- 2) A time-step and stride-oriented data segmentation scheme is developed for segment and label reshaping of raw EMG signals to facilitate better handling of large datasets and complex temporal patterns.
- 3) A processed knee-abnormalities-based HAR dataset has been formulated using raw surface electromyography (sEMG) and goniometer sensors in real-world scenarios.
- 4) A detailed analytical discussion with benchmark datasets and model comparison is carried out for a comprehensive understanding and performance evaluation of the proposed model.

The remaining sections of this article are organized as follows. Section II summarizes the state-of-the-art HAR systems, and Section III describes the proposed methodology with incorporated benchmark models and evaluation metrics. Section IV discusses the experimental results, and finally, Section V concludes the paper with future directions.

II. RELATED WORKS

Researchers [1], [15], [16], and [17] around the globe use diverse forms of learning algorithms for developing HAR systems with wearable and ambient sensors. Mekruksavanich et al. [12] proposed an efficient HAR system named *ResNeXT* with knee abnormalities using surface EMG sensors and a goniometer. They leveraged the power of the deep residual neural network for classifying human activities with automatic feature extraction using CNN.

Gautam et al. [18] proposed a transfer learning-based deep-learning model named *MyoNet* for classifying lower limb movements using the UCI EMG dataset. The proposed model is trained with knee joint angle prediction, and then the knowledge is transferred to the classification layer, incorporating the transfer learning approach.

Tokas et al. [19] proposed an effective hybrid feature engineering pipeline for spatio-temporal feature extraction on the UCI EMG dataset using parallel CNN and LSTM layers. Multiple regularization techniques were introduced on CNN layers to handle overfitting, and consecutive dense layers with ReLu and Softmax activation functions were used for adding nonlinearity and ADL classification. Similarly, Zhang et al. [20] propose a dynamic neural network algorithm based on the multifeature fusion of sEMG signals to identify eight lower limb movements accurately. The authors also introduced a novel GA-DANN recognition model that uses a genetic algorithm (GA) to optimize the back-propagation neural network (BPNN) weights. The proposed methodology improved the overall performance and achieved an average accuracy of 94.89%.

Issa and Khaled [21] propose an enhanced lower limb activity recognition system using sEMG raw data and a broad learning (BL) classifier. The proposed system calculates features by converting EMG raw data to narrow overlapping segments and finding the EMG power spectrum texture analysis. Vijayvargiya et al. [22] present a hardware-based approach for measuring and analyzing sEMG signals in the context of lower limb activity recognition. Zhang et al. [23] present a comprehensive framework for recognizing lower limb activities using sEMG physiological sensor. They combined data augmentation modalities and an enhanced capsule network (CapsNet) architecture to improve the accuracy and robustness of activity recognition. Incorporating the augmentation technique for data resampling gave the proposed model an edge over the benchmark model but certainly added a computational overhead for real-time applications.

Semwal et al. [13] address the issue of patients with walking disabilities by facilitating them with rehabilitation through accurate joint angle results in a shorter time frame. They proposed a novel joint trajectory generation using LSTM with musculoskeletal models and polynomial equations. The proposed model effectively enhances the performance of the joint angle calculations for gait analysis and rehabilitation of patients. Zhou et al. [24] proposed an enhanced deep-learning-assisted HAR model for automatic labeling and classification on weekly labeled sensor data. The presented model automates the data instance labeling using deep Q-networks, classifies ADL using LSTM with effective data preprocessing, and achieves an average AUC score of 95%. Subsequently, Wan et al. [25] propose deep-learning techniques to enhance calculation speed and recognition accuracy, aiming to reduce energy consumption and hardware cost while improving performance on multiple benchmark datasets. The proposed model demonstrates notable performance compared to the benchmark works on wearable sensor-based HAR systems.

Synthesis: The state-of-the-art benchmark works on sEMG sensor-based HAR systems introduced multiple ensembles

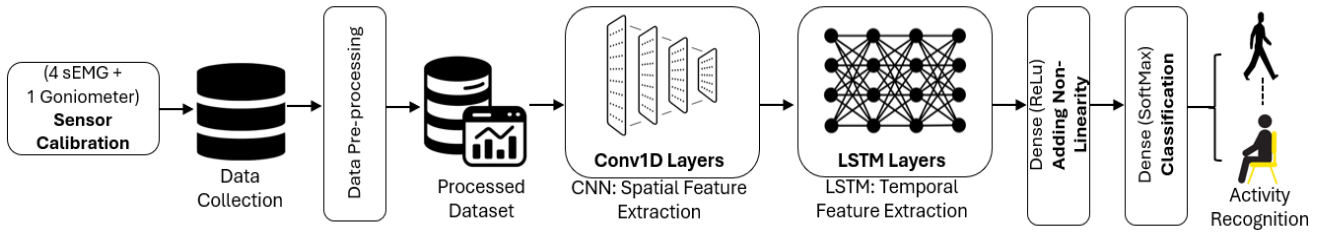


Fig. 1. Architecture of the proposed system from dataset generation to activity recognition.

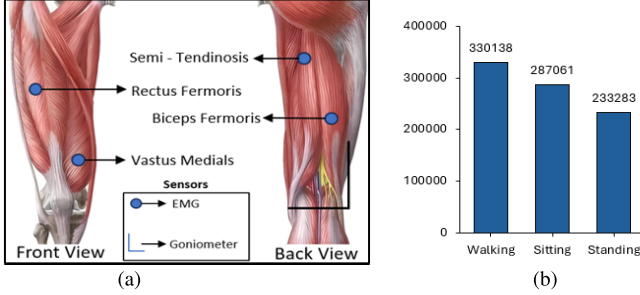


Fig. 2. Sensor mounting location and class distributions. (a) Sensor mounting location on lower limb of various subjects. (b) Activity-wise frequency count of UCI-EMG dataset.

and deep-learning models for classification and automatic feature engineering in the spatial and temporal domains for optimal performance. However, most of the benchmark's work fails to exploit the performance of different sensor fusion and lightweight deep-learning architecture for activity recognition. Additionally, in UCI-EMG-based benchmark works, researchers demonstrated the performance of the proposed model on disjoint user sets of abnormal and normal users. This hampers the overall performance and exploration of real-life activity recognition scenarios. Thus, this article introduces a novel feature fusion pipeline using hybrid deep learning and a unique time-stamp-based data segmentation scheme for physiological sensor-based HAR systems encompassing normal and abnormal user clusters.

III. PROPOSED METHODOLOGY

The methodology of the proposed model consists of multiple phases, from data collection, preparation, and pre-processing to building the feature extractor and classifier with benchmark model settings, as shown in Fig. 1.

A. Data Collection and Preparation

Multiple physiological sensor-based datasets are publicly available for medical data analysis and healthcare applications. UCI-EMG dataset for lower limb [26] is one of the prominent physiological sensor datasets generated by incorporating four channels of sEMG sensors in different portions of the knee and a goniometer—SG150B for capturing three different activities of various individuals. The sensors were mounted on the rectus femoris (RF), vastus medialis (VM), semi-tendinosus (ST), and biceps femoris (BF) of the lower limb for data collection as shown in Fig. 2(a)

$$f_s = 1000 \text{ Hz} \quad (1)$$

$$\text{Amplitude Resolution} = 2^N = 2^{14} = 16384. \quad (2)$$

Eleven subjects with knee abnormalities previously diagnosed by various healthcare professionals were considered for collecting three human activities, i.e., *Sitting*, *Standing* and *Walking*. Along with the abnormal users, 11 more subjects were considered in the dataset to make the dataset comprehensive for classification and real-life behavior, adding the total user count to 22 male users. Further, each individual's raw sensor data were collected using raw sEMG and a goniometer with the interface of MWX8 DataLog sensor by Biometrics. Raw data instances were collected using a sampling rate (f_s) and the amplitude resolution of 14-bits, providing 16384 discrete levels as described in (1) and (2), respectively. Additionally, we incorporated two benchmark public datasets named mHealth [27] and MotionSense [28], having 12 and 6 ADL, respectively, for benchmark testing and comparison of the proposed ConvLSTM model.

B. Data Preprocessing and Exploration

Once the dataset was finalized, we analyzed and processed the dataset to make it classification-ready for activity recognition. First, different users' data were combined using the pandas' library by converting the text file into comma-separated files (CSV). For combining different users' data, feature annotations were done after removing the missing instances, followed by the annotation class labels using the information directory of UCI-EMG. Then, we checked for duplicate samples and interpreted that 0.32% of data instances were redundant. We did not remove these redundant instances as in a real-time environment, multiple sensors will collect the duplicate instances, and the small percentage will not cause any bias to the model.

Further, detailed data exploration was performed on the formulated dataset, and it extrapolated the minor issue of class imbalance as shown in Fig. 2(b). Upon analyzing the class count, the minimum activity frequency was 233283 instances with *Standing* ADL. This directs us to keep the frequency count intact, as all the activity labels have many associated data patterns, as shown in Table I and will not need further data augmentation. Also, augmenting the data for this experiment will be unethical as this research can be further used for real-life knee abnormalities users for possible health and day-to-day activity monitoring for rehabilitation. After data preprocessing, the patterns associated with different attributes for normal and abnormal users with range measures are described in Table I for EMG and goniometer, respectively. Incorporating standardization or normalization of the attributes was not performed to reduce the real-time recognition overhead.

TABLE I
RANGE AND TRAJECTORIES OF THE EMG
AND GONIOMETER SENSORS

User	Sensor	Measure	Range	Trajectory
Abnormal	EMG	Min	-1.3470	
		Mean	-0.0001	
		Max	1.0860	
	Goniometer	Min	-55.700	
		Mean	-11.801	
		Max	2.2000	
Normal	EMG	Min	-0.1515	
		Mean	-0.0007	
		Max	0.1935	
	Goniometer	Min	-67.0	
		Mean	-12.574	
		Max	9.9000	

C. Proposed Model

To recognize the collected ADLs, we created novel convolutional neural networks (CNNs) with long short-term memories (LSTMs)—ConvLSTM for classifying physiological sensor-based human activities as shown in Fig. 1. Our proposed model is developed to effectively retain hierarchical features from the input data series and improve the performance of HAR systems with knee abnormalities. First, a time-distributed layer with a 1-D convolutional ($H_{\text{conv}}(t)$) operation is applied to the input data (X) for extracting local features. It uses a kernel (k) function that convolves multiple filters (F) over the input sequence and a bias (b), which handles the vanishing gradient problem as described in (3). Two successive time-distributed Conv1D layers with 16 filters and a kernel size of 2 are employed with the rectified linear unit (ReLU) activation function to introduce nonlinearity in the extracted features

$$H_{\text{conv}}(t) = \text{ReLU} \left(\sum_{k=1}^K X(t-k) * F(k) + b \right) \quad (3)$$

$$H_{\text{dropout}}(t) = \begin{cases} H_{\text{conv}}(t), & \text{with probability } 1-p \\ 0, & \text{with probability } p. \end{cases} \quad (4)$$

Further, a dropout ($H_{\text{dropout}}(t)$) layer with probability ($p = 0.1$) was applied after each Conv1D layer to enhance model generalization and prevent model overfitting as described in (4). A Max-pooling ($H_{\text{Max-pool}}(t)$) layer is also incorporated to downsample the feature maps, reducing the model's overall complexity while retaining essential features as described in (5). Subsequently, a time-distributed flatten layer (H_{flatten}) described in (6) is employed to transform the feature maps into a 1-D vector

$$H_{\text{Max-pool}}(t) = \max_{t' \in [t-W+1, t]} H_{\text{dropout}}(t') \quad (5)$$

$$H_{\text{flatten}} = [H_{\text{pool}}(1), H_{\text{pool}}(2), \dots, H_{\text{pool}}(T - W + 1)]. \quad (6)$$

Then, an LSTM layer is incorporated to capture the temporal dependencies in the sequence data. LSTM layers allow

Algorithm 1 Proposed ConvLSTM Model

```

1: Variables:  $X, W_i, b_i, d_r = 0.1$ ,  $\triangleright$  where  $X$  is the input data
   with Batch-Size ( $B_s$ ), Time Stamp ( $T$ ) and No. of Features
   ( $F$ ), weights ( $W_i$ ), bias ( $b_i$ ), and dropout ( $d_r$ ).
2: Input:  $X, W_{\text{cnn}}, b_{\text{cnn}}, d_r$   $\triangleright$  CNN parameters
3: procedure CNN( $X, W_{\text{cnn}}, b_{\text{cnn}}, d_r$ )
4:    $H_{\text{cnn}} \leftarrow \text{ReLU}(\text{Conv1D}(X, W_{\text{cnn}}, b_{\text{cnn}}))$ 
5:    $H'_{\text{cnn}} \leftarrow \text{Dropout}(H_{\text{cnn}}, d_r)$ 
6:    $H''_{\text{cnn}} \leftarrow \text{MaxPooling}(H'_{\text{cnn}}, \text{Poolsize} = 2)$ 
7:    $H'''_{\text{cnn}} \leftarrow \text{Flatten}()$ 
8:   Output:  $H'''_{\text{cnn}}$ 
9: end procedure
10: Input:  $H'''_{\text{cnn}}, W_{\text{LSTM}}, b_{\text{LSTM}}, d_r$   $\triangleright$  LSTM parameters
11: procedure LSTM( $H'''_{\text{cnn}}, W_{\text{LSTM}}, b_{\text{LSTM}}, d_r$ )
12:    $H_{\text{gru}} \leftarrow \text{LSTM}(H'''_{\text{cnn}}, W_{\text{LSTM}}, b_{\text{LSTM}})$ 
13:    $H'_{\text{LSTM}} \leftarrow \text{Dropout}(H_{\text{LSTM}}, d_r)$ 
14:   Output:  $H'_{\text{LSTM}}$ 
15: end procedure
16: Input:  $H'_{\text{LSTM}}, W_{\text{dense}}, b_{\text{dense}}, d_r, W_{\text{output}}, b_{\text{output}}$   $\triangleright$  Dense layer
   parameters
17: procedure DENSE( $H'_{\text{LSTM}}, W_{\text{dense}}, b_{\text{dense}}, d_r, W_{\text{output}}, b_{\text{output}}$ )
18:    $H_{\text{dense}} \leftarrow \text{ReLU}(\text{Dense}(H'_{\text{LSTM}}, W_{\text{dense}}, b_{\text{dense}}))$ 
19:    $H'_{\text{dense}} \leftarrow \text{Dropout}(H_{\text{dense}}, d_r)$ 
20:    $\hat{Y} \leftarrow \text{Softmax}(\text{Dense}(H'_{\text{dense}}, W_{\text{output}}, b_{\text{output}}))$ 
21:   Output:  $\hat{Y}$   $\triangleright$  Predicted class probabilities
22: end procedure

```

the model to capture long-term feature dependencies over the provided time stamps, allowing comprehensive fine-tuning of the sequential data as described in (7), where (σ) is Sigmoid and (θ) is Tanh activation function. Dropout regularization is used here as well to prevent model overfitting. Following the LSTM layer, fully connected layers were used to process the captured learnings from the above layers

$$\text{Forget Gate}(f_t) = \sigma(W_f \cdot X_t + U_f \cdot H_{t-1} + b_f) \quad (7)$$

$$\text{Input Gate}(i_t) = \sigma(W_i \cdot X_t + U_i \cdot H_{t-1} + b_i) \quad (8)$$

$$\text{Candidate}(\tilde{C}_t) = \theta(W_C \cdot X_t + U_C \cdot H_{t-1} + b_C) \quad (9)$$

$$\text{Cell State}(C_t = f_t) \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (10)$$

$$\text{Output Gate}(o_t) = \sigma(W_o \cdot X_t + U_o \cdot H_{t-1} + b_o) \quad (11)$$

$$\text{Hidden State}(H_t) = o_t \odot \tanh(C_t). \quad (12)$$

A dense layer with ReLU activation is employed with an optimal number of neuron counts to make the model computationally efficient and robust, as described in (13). Finally, the output layer consists of a dense layer with a softmax activation function to construct class probabilities for multi-class activity recognition as described in (14). Our proposed model as illustrated in Fig. 3 is compiled using *categorical cross-entropy* loss function and *Adam* optimizer for enhanced multiclass classification as described in the Algorithm 1

$$H'_t = \text{ReLU}(W_{\text{dense1}} \cdot H_t + b_{\text{dense1}}) \quad (13)$$

$$\hat{Y} = \text{Softmax}(W_{\text{output}} \cdot H'_t + b_{\text{output}}). \quad (14)$$

D. Evaluation Metrics

Along with the proposed model, multiple ensemble methods such as random forest (RF), eXtreme gradient boosting (xGB), and deep-learning models like—ANN, LSTM, and CNN-GRU were incorporated for detailed analysis and comparison with

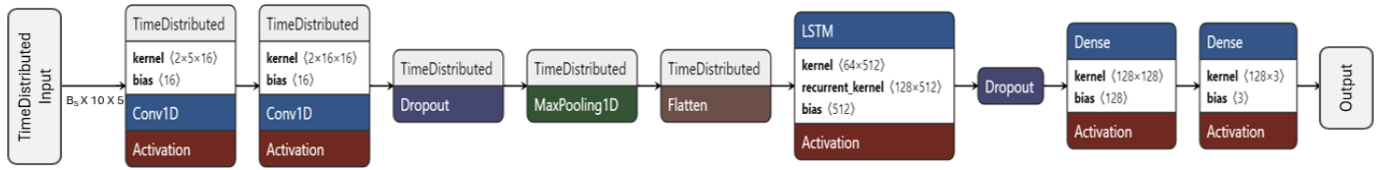


Fig. 3. Network architecture of the proposed ConvLSTM model.

TABLE II
DETAILED PERFORMANCE COMPARISON OF THE PROPOSED MODEL WITH INCORPORATED BENCHMARK MODELS

Models / Metrics	High. Acc. (%)	5-Fold Avg. Acc.(%)	Avg. Mt.	P	R	F1	Testing Loss	C_t in Frac. Sec.
xGB	76.86	76	M.A.	0.78	0.76	0.76	0.524 ± 0.538	550 ± 672
			W.A.	0.77	0.77	0.77		
RF	79.04	78	M.A.	0.79	0.79	0.79	0.511 ± 0.530	189 ± 241
			W.A.	0.79	0.79	0.79		
ANN	87.21	87	M.A.	0.87	0.87	0.87	0.441 ± 0.474	718 ± 769
			W.A.	0.87	0.87	0.87		
LSTM	96.52	96	M.A.	0.96	0.96	0.96	0.0907 ± 0.098	3850 ± 3955
			W.A.	0.97	0.97	0.97		
CNN-GRU	98.60	98	M.A.	0.99	0.99	0.99	0.0421 ± 0.0441	600 ± 661
			W.A.	0.99	0.99	0.99		
Proposed ConvLSTM	99.21	99	M.A.	0.99	0.99	0.99	0.0240 ± 0.0258	330 ± 342
			W.A.	0.99	0.99	0.99		

* High. Acc = Highest Accuracy, Avg. Acc. = Average Accuracy, Avg. Mt. = Averaging Method, M.A. = Macro Average, W.A. = Weighted Average, C_t = Computational time

the proposed model. These models were chosen because their learning paradigms differ and perform efficiently toward sensor-based HAR system [5], [12], [29]. Also, we implemented the benchmark models using the same hyperparameters as our proposed ConvLSTM model to maintain a consistent comparison framework for evaluation. Further, to test and benchmark the proposed model, we incorporated multiple evaluation metrics like—precision (P), recall (R), $F1$ -score ($F1$), accuracy (Acc.), computational time (C_t) = ($C_{tr} + C_{val}$), and performance loss for detailed performance comparison [17], [30].

IV. EXPERIMENTAL RESULTS AND DISCUSSION

We used Python (version—3.9.7) programming language to develop our model and incorporated TensorFlow (version—2.6.0) with Keras (version—2.6.0) and Sk-learn as the primary libraries for developing the proposed and benchmark models. The model training, testing, and validation were conducted on a Windows-based dedicated hardware machine having 32 GB of RAM, 2 TB HDD, an Intel Xeon W-2134 processor, and a NVIDIA Quadro P2000 GPU.

A. Performance of the Proposed ConvLSTM Model

To train our proposed model with the processed data, we segmented the dataset based on the time steps (t_s), feature-length (n_f), and step size (step) as described in the Algorithm 2. An adequate batch size of 256 for 100 training epochs with a neuron count of 128 for LSTM and dense layers was used to lower the computational overhead during training and validation. Other incorporated hyperparameters for the proposed model are described in the Algorithm 1. Upon testing, the proposed model achieved the highest accuracy of 99.21% and an average accuracy of 99% after fivefold cross-validation. Other performance metrics, such as precision, recall, and $F1$ -score, were also efficient and surpassed the

Algorithm 2 Stride and Time Step-Based Instance Segmentation Algorithm

```

1: Input: Dataset ( $df$ ), Time steps ( $t_s$ ), Step size ( $step$ ), Number of features ( $n_f$ ) = 5;
2: Initializing Empty Lists:  $segments = []$ ,  $labels = []$ ;
3: for  $[i \text{ in range}(0, \text{len}(df) - t_s, step)]$  do
4:    $f1 = df[EMG1].values[i : i + t_s]$ ;
5:    $f2 = df[EMG2].values[i : i + t_s]$ ;
6:    $f3 = df[EMG3].values[i : i + t_s]$ ;
7:    $f4 = df[EMG4].values[i : i + t_s]$ ;
8:    $f5 = df[Goniometer].values[i : i + t_s]$ ;
9:    $label = \text{mode}(df[Class][i : i + t_s])$ ;
10:   $segments.append([f1, f2, f3, f4, f5])$ ;
11:   $labels.append(label)$ ;
12: end for
13: Output:  $segments, labels$ ;

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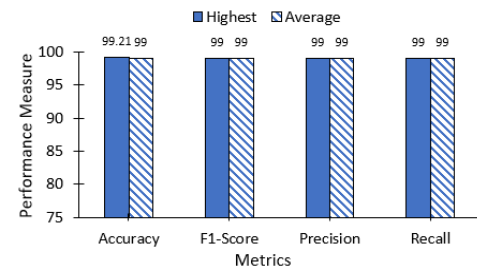


Fig. 4. Performance comparison of the proposed ConvLSTM model using fivefold validation.

incorporated benchmark models with an optimal performance margin. The maximum and average performances of the proposed model are illustrated in Fig. 4. Also, the detailed performance comparison of the proposed model with benchmarks is described in Table II.

In Section III-B, we addressed a minor class imbalance in the dataset. The proposed model effectively tackled this issue

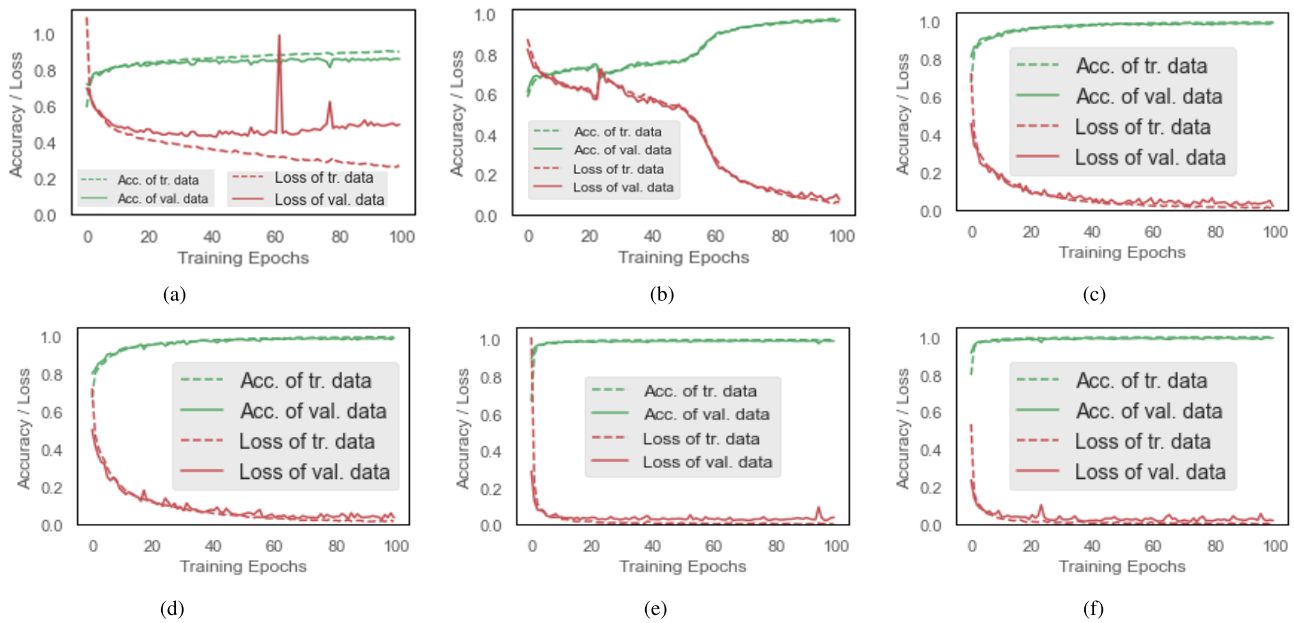


Fig. 5. Accuracy and loss plots of the different deep-learning models and the proposed ConvLSTM model. (a) Accuracy and loss plot of ANN classifier. (b) Accuracy and loss plot of LSTM. (c) Accuracy and loss plot of CNN-GRU. (d) Accuracy and loss plot of the proposed ConvLSTM. (e) Accuracy and loss plot of the proposed ConvLSTM on mHealth. (f) Accuracy and loss plot of the proposed ConvLSTM on MotionSense.

as the extracted features from the convolution and LSTM layers were robust for all the classes. It is also evident through the proposed model's nonvolatile training and validation losses, as illustrated in Fig. 5(d). The proposed model inherently extracts hierarchical features in different layers, allowing the model to learn convoluted feature sets for classifying intricate data patterns. Moreover, the sequential nature of LSTM layers allows the model to filter out the noisy data over time. The noisy minority or majority class instances can be filtered out using LSTM layers, allowing the proposed model to remove irrelevant data patterns and enhancing the model's focus on important class-related features.

On the contrary, we can see that nonhybrid benchmark models like—ANN and LSTM suffer through high training and validation loss, as shown in Fig. 5(a) and (b), respectively, due to the absence of spatial feature engineering by convolution layers. ANN needs manual feature extraction for the multidimensional features-oriented datasets to perform optimally, which is not friendly toward computational time and human resources. Also, LSTMs are efficient for sequential data, but their performance heavily depends on the quality of the features fed into them. As our features have raw physiological sensor data, LSTM struggled to generalize the model in the provided training time, as illustrated in Fig. 5(b). Moreover, CNN-GRU managed to generalize the data patterns optimally since the training and validation losses were minimal, as shown in Fig. 5(c). However, the model overfits toward the dominant data patterns of the majority class, ignoring the minority patterns.

B. Efficacy Analysis With the Benchmark Models and Datasets

In Section III-C, we included five machine- and deep-learning models in our experiment to compare and analyze

the performance of the proposed model. The same hyper-parameters were used for all the deep-learning benchmark models, and default sklearn's parameters were incorporated for xGB and RF classifiers. Upon classifying ADLs using ensemble approaches like—RF and xGB, the overall classification performance decreased significantly as they could not optimally generalize the complex raw instances. xGB achieved an average accuracy of 76%, while RF achieved 78%. This decline in performance is because of the absence of automatic feature engineering and issues related to class imbalance, which negatively impact model generalization. The detailed performance achieved by the ensemble models is described in Table II.

Further, ANN and LSTM surpassed ensemble models in performance but need spatial feature analysis for better model generalization to cope with hybrid learning approaches. LSTM takes the highest computational time of 3955 fs as the input sequences are long and the dataset has intricate temporal patterns. The proposed model demonstrates faster computational times than the benchmark models due to the inclusion of multiple regularization techniques and parallel processing of convolution layers, resulting in a sparser model and reduced overfitting as described in Table II. Also, the incorporation of ReLU in the spatial feature extraction phase made the feature set minimal for temporal feature engineering, making the hybrid deep-learning models (proposed ConvLSTM and CNN-GRU) overcome the vanishing gradient problem and avoid saturation, allowing them to learn faster. Compared to the CNN-GRU, the proposed ConvLSTM model effectively generalized the data patterns in comparatively faster computational time without model overfitting. Additional reset and update gates in the CNN-GRU model introduce complex weight updates, leading the model toward inefficient training capabilities. The proposed model enhances

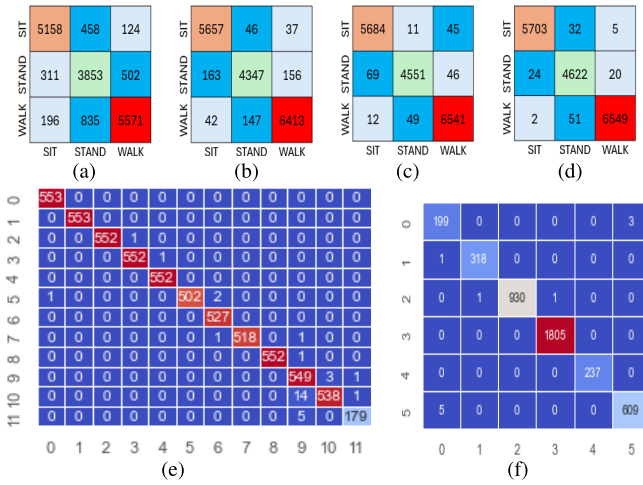


Fig. 6. Confusion matrix of the proposed ConvLSTM model with benchmarks on diverse datasets. (a) ANN on UCI-EMG. (b) LSTM on UCI-EMG. (c) CNN-GRU on UCI-EMG. (d) Proposed model on UCI-EMG. (e) Proposed model on mHealth. (f) Proposed Model on Motion-Sense.

TABLE III
PERFORMANCE COMPARISON OF THE PROPOSED MODEL
WITH THE BENCHMARK PUBLIC DATASETS

Dataset	# ADL	Acc.	5-Fold Acc.	Avg. Mt.	P	R	F1
mHealth [27]	12	99.34	99.11	M.A.	0.99	0.99	0.99
				W.A.	0.99	0.99	0.99
Motion Sense [28]	6	99.52	99.32	M.A.	0.99	0.99	0.99
				W.A.	1	1	1

the computational times by 87.03%, 168.16%, and 63.60% concerning ANN, LSTM, and CNN-GRU, respectively. Subsequently, Fig. 6 displays the confusion matrix of our proposed model, i.e., Fig. 6(d), highlighting higher true positive counts compared to benchmark deep-learning models like—ANN, LSTM, and ConvLSTM models as shown in Fig. 6(a)–(c), respectively. The absence of convoluted spatial features significantly compromised the performance of both ensemble and nonhybrid deep learning models (ANN, LSTM), leading to a considerable increase in Type-1 and Type-2 errors.

Additionally, to validate the generalization capability of the proposed model, we incorporated additional public datasets comprising various static and dynamic activities. The proposed ConvLSTM model demonstrated its ability to generalize well on diverse public datasets beyond the initial experiments, as described in Table III achieving an optimal recognition performance of 99.34% on the mHealth [27] dataset with 12 ADLs. Similarly, on the MotionSense [28] dataset with six ADL, the proposed model achieved the highest recognition accuracy of 99.52%. The consistent fivefold cross-validation results of 99.11% and 99.32% on these respective datasets further validate the model's strong generalization capabilities. Further, the model's training loss and error rates were notably low, as illustrated in Figs. 5 and 6, underscoring its ability to effectively learn and extract both spatial and temporal features from complex, multiactivity benchmark datasets.

TABLE IV
STATE-OF-THE-ART BENCHMARK PERFORMANCE COMPARISON
OF THE PROPOSED MODEL WITH BENCHMARKS

Benchmark Work	Year	Acc.	F1	Loss
Naik <i>et al.</i> [32]	2018	87.63%	88%	—
Gautam <i>et al.</i> [18]	2020	95.2%	—	—
Vijayvargiya <i>et al.</i> [33]	2021	97.45%	—	—
Issa <i>et al.</i> [21]	2022	94%	—	—
Vijayvargiya <i>et al.</i> [34]	2022	99.15%	99%	—
Mekruksavanich <i>et al.</i> [12]	2023	93.52%	93.31%	0.27
Vijayvargiya <i>et al.</i> [31]	2023	98.50%	99%	—
Tokas <i>et al.</i> [19]	2024	98.61%	99	—
Proposed ConvLSTM Model	2024	99.21%	99%	0.0249

C. State-of-the-Art Model Comparisons

To contextually evaluate and validate the proposed model, we compared the model's performance with multiple state-of-the-art benchmark HAR systems on the same UCIEMG dataset. The proposed model outperformed the benchmark models with optimal performance margin as described in Table IV. The model achieved outstanding accuracy and a significantly lesser loss value of 0.0249. These results demonstrate our approach's efficacy in accurately classifying EMG signals, surpassing all previous benchmarks. Moreover, the most recent state-of-the-art HAR model on the UCIEMG dataset achieved an average performance accuracy of 98.50% [31] and 93.52% [12]. The proposed KEE-HAR model managed to achieve an average performance accuracy of 99.21%, surpassing the previous benchmark works as described in the Table IV.

V. CONCLUSION AND FUTURE WORK

In this paper, we design and develop an efficient hybrid deep-learning model (ConvLSTM) for recognizing individual daily living activities with knee abnormalities. The data were collected using four sEMG physiological sensors mounted on different muscles of lower limb areas with a goniometer sensor for perceiving knee rotation. An adequate data preprocessing pipeline is also utilized to make the dataset robust in terms of missing data, duplicates, and attribute interpretation. The proposed model achieved the highest and average accuracy of 99.21% and 99%, respectively, with the class imbalance problem intact, demonstrating the efficiency of our proposed model's spatial and temporal feature handling. Further, the proposed model outperformed all the incorporated benchmark and state-of-the-art models with optimal performance margins and demonstrated its capability of recognizing human activities with knee abnormalities. Moreover, the computational times were also efficient for model generalization and managed to optimize the computational times by 87.03%, 168.16%, and 63.60% concerning ANN, LSTM, and CNN-GRU, respectively. Furthermore, as the limitation of the work presented in this article, the proposed model lacks real-time testing in the medical environment, which will endure its robustness and efficiency.

In the future, we plan to develop a streamlined deep learning-based activity recognition pipeline using physiological sensors for critical care. To expand our model's

adaptability, we will explore additional biosignal processing techniques, such as tunable-Q wavelet transform (TQWT)-based sub-band separation [35] and advanced time-frequency analysis methods [36] for feature engineering toward diverse healthcare scenarios.

CODE AND DATA AVAILABILITY

Dataset is publicly available at <http://archive.ics.uci.edu/dataset/278/emg+dataset+in+lower+limb>. Code (in Python) will be provided for academic and research purposes on reasonable request.

REFERENCES

- [1] V. B. Semwal et al., "Pattern identification of different human joints for different human walking styles using inertial measurement unit (IMU) sensor," *Artif. Intell. Rev.*, vol. 55, no. 2, pp. 1149–1169, 2022.
- [2] T. R. Mim et al., "GRU-INC: An inception-attention based approach using GRU for human activity recognition," *Expert Syst. Appl.*, vol. 216, Apr. 2023, Art. no. 119419.
- [3] E. Ramanujam, T. Perumal, and S. Padmavathi, "Human activity recognition with smartphone and wearable sensors using deep learning techniques: A review," *IEEE Sensors J.*, vol. 21, no. 12, pp. 13029–13040, Jun. 2021.
- [4] S. K. Challa, A. Kumar, V. B. Semwal, and N. Dua, "An optimized deep learning model for human activity recognition using inertial measurement units," *Expert Syst.*, vol. 40, no. 10, p. 13457, Dec. 2023.
- [5] N. A. Choudhury and B. Soni, "An adaptive batch size-based-CNN-LSTM framework for human activity recognition in uncontrolled environment," *IEEE Trans. Ind. Informat.*, vol. 19, no. 10, pp. 10379–10387, Oct. 2023.
- [6] M. Bisoi, B. K. Balabantaray, and S. Moulik, "Human activity recognition using CTAL model," in *Computer Vision and Machine Intelligence*, M. Tistarelli, S. R. Dubey, S. K. Singh, and X. Jiang, Eds., Singapore: Springer, 2023, pp. 347–356.
- [7] V. B. Semwal, A. Gupta, and P. Lalwani, "An optimized hybrid deep learning model using ensemble learning approach for human walking activities recognition," *J. Supercomput.*, vol. 77, no. 11, pp. 12256–12279, Nov. 2021.
- [8] N. Gaud, M. Rathore, and U. Suman, "Hybrid deep learning-based human activity recognition (har) using wearable sensors: An edge computing approach," in *Proceedings of Data Analytics and Management*, A. Swaroop, Z. Polkowski, S. D. Correia, and B. Virdee, Eds., Singapore: Springer, 2024, pp. 399–410.
- [9] K. K. Makam, V. K. Singh, and R. B. Pachori, "ALS detection framework based on automatic singular spectrum analysis and quantum convolutional neural network from EMG signals," *IEEE Sensors Lett.*, vol. 8, no. 9, pp. 1–4, Sep. 2024.
- [10] A. Pradhan and S. Srivastava, "Hierarchical extreme puzzle learning machine-based emotion recognition using multimodal physiological signals," *Biomed. Signal Process. Control*, vol. 83, May 2023, Art. no. 104624.
- [11] H. Swierczynski, J. Pukacki, S. Szczesny, C. Mazurek, and R. Wasilewicz, "Sensor data analysis and development of machine learning models for detection of glaucoma," *Biomed. Signal Process. Control*, vol. 86, Sep. 2023, Art. no. 105350.
- [12] S. Mekruksavanich, P. Jantawong, N. Hnoohom, and A. Jitpattanakul, "Human activity recognition for people with knee abnormality using surface electromyography and knee angle sensors," in *Proc. Joint Int. Conf. Digit. Arts, Media Technol. ECTI Northern Sect. Conf. Electr., Electron., Comput. Telecommun. Eng. (ECTI DAMT NCON)*, Mar. 2023, pp. 483–487.
- [13] V. B. Semwal et al., "Development of the LSTM model and universal polynomial equation for all the sub-phases of human gait," *IEEE Sensors J.*, vol. 23, no. 14, pp. 15892–15900, Jul. 2023.
- [14] R. R. Sharma, M. Kumar, and R. B. Pachori, "Classification of EMG signals using eigenvalue decomposition-based time-frequency representation," in *Biomedical and Clinical Engineering for Healthcare Advancement*. Hershey, PA, USA: IGI Global, 2020, pp. 96–118.
- [15] N. A. Choudhury and B. Soni, "In-depth analysis of design & development for sensor-based human activity recognition system," *Multimedia Tools Appl.*, vol. 83, no. 29, pp. 73233–73272, Sep. 2023.
- [16] S. Moulik and S. Majumdar, "FallSense: An automatic fall detection and alarm generation system in IoT-enabled environment," *IEEE Sensors J.*, vol. 19, no. 19, pp. 8452–8459, Oct. 2019.
- [17] N. A. Choudhury and B. Soni, "Enhanced complex human activity recognition system: A proficient deep learning framework exploiting physiological sensors and feature learning," *IEEE Sensors Lett.*, vol. 7, no. 11, pp. 1–4, Nov. 2023.
- [18] A. Gautam, M. Panwar, D. Biswas, and A. Acharyya, "MyoNet: A transfer-learning-based LRCN for lower limb movement recognition and knee joint angle prediction for remote monitoring of rehabilitation progress from sEMG," *IEEE J. Transl. Eng. Health Med.*, vol. 8, pp. 1–10, 2020.
- [19] P. Tokas, V. B. Semwal, and S. Jain, "Deep ensemble learning approach for lower limb movement recognition from multichannel sEMG signals," *Neural Comput. Appl.*, vol. 36, no. 13, pp. 7373–7388, May 2024.
- [20] P. Zhang, J. Zhang, and A. Elsabbagh, "Lower limb motion intention recognition based on sEMG fusion features," *IEEE Sensors J.*, vol. 22, no. 7, pp. 7005–7014, Apr. 2022.
- [21] S. Issa and A. R. Khaled, "Lower limb activity prediction using emg signals and broad learning," *Int. J. Comput. Inf. Syst. Ind. Manag. Appl.*, vol. 14, p. 11, Jan. 2022.
- [22] A. Vijayvargiya, P. Singh, R. Kumar, and N. Dey, "Hardware implementation for lower limb surface EMG measurement and analysis using explainable AI for activity recognition," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–9, 2022.
- [23] C. Zhang, Y. Li, Z. Yu, X. Huang, J. Xu, and C. Deng, "An end-to-end lower limb activity recognition framework based on sEMG data augmentation and enhanced CapsNet," *Expert Syst. Appl.*, vol. 227, Oct. 2023, Art. no. 120257.
- [24] X. Zhou, W. Liang, K. I. Wang, H. Wang, L. T. Yang, and Q. Jin, "Deep-learning-enhanced human activity recognition for Internet of Healthcare Things," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 6429–6438, Jul. 2020.
- [25] S. Wan, L. Qi, X. Xu, C. Tong, and Z. Gu, "Deep learning models for real-time human activity recognition with smartphones," *Mobile Netw. Appl.*, vol. 25, no. 2, pp. 743–755, Apr. 2020.
- [26] O. Sanchez and J. Sotelo, 2014, "EMG dataset in lower limb," UCI Mach. Learn. Repository, doi: [10.24432/C5ZW3P](https://doi.org/10.24432/C5ZW3P).
- [27] O. Banos et al., "mhealthdroid: A novel framework for agile development of mobile health applications," in *Ambient Assisted Living and Daily Activities*, L. Pecchia, L. L. Chen, C. Nugent, and J. Bravo, Eds., Cham, Switzerland: Springer, 2014, pp. 91–98.
- [28] M. Malekzadeh, R. G. Clegg, A. Cavallaro, and H. Haddadi, "Mobile sensor data anonymization," in *Proc. Int. Conf. Internet Things Design Implement.*, New York, NY, USA, Apr. 2019, pp. 49–58.
- [29] S. Mekruksavanich and A. Jitpattanakul, "LSTM networks using smartphone data for sensor-based human activity recognition in smart homes," *Sensors*, vol. 21, no. 5, p. 1636, Feb. 2021.
- [30] N. A. Choudhury and B. Soni, "An efficient and lightweight deep learning model for human activity recognition on raw sensor data in uncontrolled environment," *IEEE Sensors J.*, vol. 23, no. 20, pp. 25579–25586, Oct. 2023.
- [31] A. Vijayvargiya, R. Kumar, and P. Sharma, "PC-GNN: Pearson correlation-based graph neural network for recognition of human lower limb activity using sEMG signal," *IEEE Trans. Human-Mach. Syst.*, vol. 53, no. 6, pp. 1–10, Dec. 2023.
- [32] G. R. Naik et al., "An ICA-EBM-based sEMG classifier for recognizing lower limb movements in individuals with and without knee pathology," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 26, no. 3, pp. 675–686, Mar. 2018.
- [33] A. Vijayvargiya, V. Gupta, R. Kumar, N. Dey, and J. M. R. S. Tavares, "A hybrid WD-EEMD sEMG feature extraction technique for lower limb activity recognition," *IEEE Sensors J.*, vol. 21, no. 18, pp. 20431–20439, Sep. 2021.
- [34] A. Vijayvargiya, B. Singh, R. Kumar, U. Desai, and J. Hemanth, "Hybrid deep learning approaches for sEMG signal-based lower limb activity recognition," *Math. Problems Eng.*, vol. 2022, pp. 1–12, Nov. 2022.
- [35] D. Pachori and T. K. Gandhi, "Automated human emotion recognition system using TQWT-based EEG subbands," *IEEE Sensors Lett.*, vol. 8, no. 12, pp. 1–4, Dec. 2024.
- [36] R. B. Pachori, *Time-Frequency Analysis Techniques and their Applications*. Boca Raton, FL, USA: CRC Press, Mar. 2023.